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What are the medium-term educational and labour market effects of private schooling?

Léonard Moulin 

Institut national d'études démographiques (INED), Aubervilliers, France

ABSTRACT

This article examines the impact of private school attendance on a range of outcomes during adulthood. Students are followed for seventeen years after they enter the sixth grade. The effect of private schooling on educational and labour market outcomes is estimated using propensity score matching. Private schooling has a positive effect on the likelihood of obtaining the *baccalauréat* and a higher education qualification, as well as obtaining a higher wage, for both girls and boys. A series of robustness tests confirms these results.

KEYWORDS

Private schools; educational outcomes; labour market; medium-term effects; propensity score matching

JEL CLASSIFICATION

I21; I24; I26

1. Introduction

Since the report of Coleman, Hoffer, and Kilgore (1982), there has been a burgeoning literature on the performance of private schools. Many studies have looked at the short-term educational effects of private schooling. For example, using the lottery admission system to control for the non-random selection of the students in charter schools in the United States, several studies have highlighted a positive impact on enrolment in a charter school on student achievement measured by gains in test scores (Abdulkadiroğlu et al. 2011; Angrist, Pathak, and Walters 2013; Bifulco and Ladd 2006; Dobbie and Fryer 2013; Gleason et al. 2014; Sass 2006). However, fewer studies have addressed medium-term effects of private schooling. This is principally because establishing estimates of longer-term variables such as college completion and earnings requires a cohort followed during the full school trajectory and beyond, into the labour market (Angrist et al. 2016).

Nevertheless, beyond the short-term educational effects of private school attendance, it is important to take a broader view, and examining the effects of private schooling both on educational outcomes in the years after lower secondary schooling and labour market outcomes.¹ This promises to bring us closer to capturing the full range of effects of enrolling in

a private school for students. Examining these effects is crucial, however, as they represent important contributions to the debate on the funding of private education. As Brown and Belfield (2001) explains, understanding some non-educational effects of private schooling, notably on wages, is important for the equity and efficiency of education systems, perhaps even more so than their effects on test scores.

In this paper, I estimate the impact of attending a private school on a range of outcomes during adulthood in France. French lower secondary private schools have the particularity of being mostly financed by the state. They are similar to government-dependent private schools, such as charter schools in the United States or free schools in England. To do this, I use data from a panel of students who are followed throughout their secondary schooling, higher education and early working life. Using this detailed dataset, I am able to measure the impact of private schooling on higher educational outcomes up to twelve years after the end of lower secondary education and on labour market outcomes up to thirteen years after. In the absence of natural randomization of students to private and public schools (through a lottery mechanism for example), my estimation strategy is based on propensity score

CONTACT Léonard Moulin  leonard.moulin@ined.fr  Institut national d'études démographiques (INED), F-93300 Aubervilliers, France

¹What I refer to in this article as medium-term impacts.

matching.² This method allows me to control for the observed reasons why students choose to attend private schools. Thus, conditionally on the propensity score, each public school student has the same probability of assignment to private school as one of the private school students, as in a randomized experiment (Dehejia and Wahba 2002). As the matching cannot eliminate possibilities of having unobservable confounders, I assess the influence of potential unobserved heterogeneity using simulation approach (Nannicini 2007; Ichino, Mealli, and Nannicini 2008).

The results on educational outcomes show that private school attendance increases the likelihood of obtaining the *baccalauréat* and a higher education qualification for both boys and girls. Moreover, I find a positive and significant effect of private schooling on boys and girls' earnings at age 27. A series of robustness tests confirm the robustness of these results.

The remainder of the paper is structured as follows. Section II presents theoretical considerations and previous evidence. Section III provides background on the French education system and describes the data used here. Section IV presents the empirical strategy. Section V presents the results and robustness analyses. Section VI concludes.

II. Theoretical considerations and previous evidence

Much of the literature on the medium-term effects of publicly funded private schooling has focused on educational outcomes. Using admission lotteries to account the non-random selection of the students enrolled in Noble Street Charter School, a high-performing charter high school in Chicago, Davis and Heller (2019) highlight that seven to nine years after the lottery, the winners are more likely to attend college, to persist in college, and enrol at selective four-year institutions. Using data from the Promise Academy in the Harlem Children's Zone, and the same methodology, Dobbie and Fryer (2015) also find a positive effect of attendance at a high-performing charter school on immediate

college enrolment and four-year college attendance. In concert with Davis and Heller (2019), Dobbie and Fryer (2015), using admissions lotteries to estimate the effects of charter school attendance on medium-term educational outcomes, Angrist et al. (2016) show that charter high school attendance in the Boston area induces a shift from two-year to four-year colleges. Using an estimation strategy that compares students enrolled in charter schools in ninth grade with students who switched to conventional public schools, Sass et al. (2016) find that students who attend a charter high school are more likely to graduate from high school and to enrol in college. Using a similar research design, Booker et al. (2011) estimate that charter high school students are more likely to earn a standard diploma and to attend college. To explain the medium-term effects of attending charter schools, Dobbie and Fryer (2015) evoke the idea that these schools may be effective at implementing educational 'best practices' which enable students to develop skills that lead not only to short-term educational gains, but also to medium-term gains.

Within the literature on the medium-term effects of private schooling, some studies have also focused on the wage gains associated with publicly funded private schooling. In Florida, Sass et al. (2016) show that students who attended a charter high school have higher wages in their mid-twenties. They argue that the medium-term gains of charter school students can be explained by the possibility that charter high schools endow students with abilities (e.g. grit, persistence, self-control, and conscientiousness) which are useful for college and career success. In contrast, Dobbie and Fryer (2020) found a negative impact of charter schools on earnings in Texas. The authors explain that this effect could be related to the fact that charter students seem to work in lower-paying industries.

III. Institutional background, data and summary statistics

French education system

In France, education is compulsory between the ages of 3³ and 16 years and students are full-time

²In particular, propensity score matching has been used in various studies examining the effect of enrolment in non-government private schools on test scores in several countries (see Moulin 2022a).

students who attend school all day long. Before the *baccalauréat*, the French national exam at the end of secondary education, the school system is divided into four major levels: *maternelle* (kindergarten), *primaire* (primary school: first to fifth grade), *collège* (lower secondary school: sixth to ninth grade), and *lycée* (upper secondary education: tenth to twelfth grade). This is the scheme for the general curriculum. There are also parallel systems following lower secondary school for technical or vocational education. At the end of secondary school, students may choose to continue their studies in higher education. There are four main academic sectors students can access after graduating from high school. In one track, students can attend *classes préparatoires*: classes/schools whose aim is to prepare students for competitive national examinations, to attempt to win entry into one of the various *grandes écoles*. The other three sectors are universities, vocational schools (in art or architecture, for example), and secondary school vocational diplomas.

One fifth (21.5%) of lower secondary students in France in 2019 were enrolled in private schools (author's calculations from MENJ - DEPP 2020). The vast majority (98%) were enrolled in a private school under contract with the state (*écoles privées sous contrat*) (MENJ - DEPP 2020). In this study, I only consider this type of private school. The majority of students enrolled in a private school attend a Catholic school (97% of them – author's calculations from MENJ - DEPP 2021; SGEC 2021). Nevertheless, religious education has almost disappeared from these schools (Langouët and Léger 1997; Valdenaire 2011) and is no longer a reason to send children to a private school. Today, the choice of private school is driven by educational motivations, and more specifically by the motivation of avoiding the constraints of strict

school zoning (Fack and Grenet 2010)– in relation to the quality of local public and private schools. As mentioned by Valdenaire (2011), private schools in France receive government funding and are strictly regulated by the state. Curricula, pedagogical objectives, and academic standards are the same in public and private schools. The only distinct form of flexibility that private schools enjoy with respect to public schools is the free recruitment of both teachers⁴ and students.⁵ Because the number of private schools is limited by the government, not all students whose families wish to enrol at them are able to do so, and there is student selection. The rules for admission to a private school can vary between institutions (Fack and Grenet 2010), but the most commonly used criteria are the student's academic level and motivation to enter the school – as assessed via a cover letter and/or an interview (Boutchénik, Givord, and Monso 2021).

Data

In this paper I use data from the DEPP-EVA⁶ panel administered by the French Ministry of Education and the French National Institute of Statistics and Economic Studies. The initial survey, the *panel d'élèves du second degré, recrutement 1995*⁷ [panel of secondary school students, 1995 recruitment], follows the educational trajectories of 17,830 students who entered sixth grade in 1995. The students were followed throughout their secondary schooling.⁸ After leaving school or after the *baccalauréat*, these young people were followed by the EVA⁹ and SUP surveys,¹⁰ which collect information every year on higher education, job market integration, and family situation until 2012. Data on higher education were collected for nine years after the *baccalauréat* and data on labour market and family situations are reported

³Following a law enacted in 2019, all children aged 3, 4 and 5 are subject to compulsory education.

⁴Teachers in both public schools and private schools under contract with the state are recruited through national competitive examinations. Public school teachers are civil servants, and are assigned to a school through a regional administrative procedure. Private school teachers are recruited by the school itself.

⁵Students are assigned to a public school on the basis of their area of residence. Private schools under contract recruit their students themselves according to their own criteria.

⁶DEPP: Direction de l'évaluation, de la prospective et de la performance [Department for evaluation, forecasting, and performance]; EVA: Entrée dans la vie adulte [Entry into adult life].

⁷Panel d'élèves du second degré, recrutement 1995 - 1995-2006 - (2006) [electronic file], DEPP [producer], Centre Maurice Halbwachs (CMH) [distributor].

⁸An *a posteriori* reconstruction of their schooling in primary education is also available.

⁹Enquête sur l'entrée dans la vie adulte des élèves entrés en 6ème en 1995 (EVA) - 2005-2012 - (2012) [electronic file], INSEE [producer], ADISP [distributor].

¹⁰When students leave secondary education and enter higher education they are included in a specific survey system, SUP (*Suivi des bacheliers dans le supérieur*, or Follow-up of *baccalauréat* holders in higher education), before being integrated into the EVA survey system after completing higher education.

during ten years after the *baccalauréat*. In addition, the system included occasional surveys that provide additional data: a family survey in 1998 that examined parents' representations and practices, and a youth survey in 2002 on career plans, possible higher education plans, and self-image. This very complete system of surveys can be used to follow young people up to seventeen years after their entry into sixth grade, and to observe medium-term outcomes in detail (Canaan and Mouganie 2018). The disadvantage of such a long follow-up time is high attrition, as some young people move abroad, cannot be reached, or refuse to respond. Several weightings have been developed to take this attrition into account.¹¹ To estimate treatment effects on each set of outcomes, I use the associated population survey weights.^{12,13}

My analysis includes three dependent variables grouped into two categories: educational outcomes (obtaining the *baccalauréat*, obtaining a higher education qualification), and labour market outcomes (monthly wage). The *baccalauréat* variable is defined as holding a general, technical or vocational *baccalauréat*, regardless of the year in which it was obtained. The higher education variable is an indicator of whether the highest diploma obtained by the student is a higher education qualification (with a minimum level of two years of study after the *baccalauréat*). To construct this variable, I consider the student's graduation from higher education at any time between 2004 and 2011. Finally, I consider the monthly salary reported by students in 2012. Thus, my sample comprises one set of observations per student. For each student I am interested in having the *baccalauréat*, not in the year in which he or she obtained it, in having a higher education diploma, whatever the diploma obtained and the year of obtaining it, as well as in his or

her monthly salary in 2012. Each outcome for each student is therefore only observed once and not repeatedly over time.¹⁴

I restrict my sample to students who were enrolled continuously in public or private school between the sixth and ninth grades. I make this choice because switchers differ from students enrolled only in public or private school (Caille 2004) and they are a small and unrepresentative population (Spees and Lauen 2019). Thus, students who change school between the sixth and ninth grades are only taken into account in my estimation strategy if they have changed school within the same school sector, preventing the possible contamination of the treated and/or control group by students who were enrolled in both private and public schools during their lower secondary education. Moreover, to make sure that I have some sociodemographic characteristics on all students, I only consider students who participated in the 1998 family survey.¹⁵ Moreover, I also exclude students whose record does not include a score for the standardized sixth grade entrance examination,¹⁶ as this variable is used later in the calculation of the propensity score. Finally, I do not consider students who were enrolled in a non-academic track in the sixth grade.¹⁷ My final sample comprises 12,572 students.

Summary statistics

Table 1 presents summary statistics for boys and girls enrolled in private schools (treatment group) and public schools (control group). There are significant differences between the two school types in the distribution of sociodemographic characteristics. My analysis includes a series of variables, which are divided into three categories: sociodemographic characteristics, school characteristics, and students'

¹¹ However, attrition is not independent of private schooling: it is more frequent among public school students.

¹² The weights used depend on the survey from which the outcome of interest is drawn. The weights of the family survey are applied to the outcome 'obtaining a *baccalauréat*', and those of the EVA survey are applied to the outcomes 'higher education qualification', and 'monthly wages' at age 27.

¹³ Using unweighted regressions produces unchanged matching estimation results.

¹⁴ This latter characteristic of my sample does not allow the use of year or student fixed effects. Similarly, as my sample is a representative panel of French students, I have on average less than 3 students per school, not allowing the use of school fixed effects.

¹⁵ There is no significant difference between public (87.3%) and private (88.7%) school students in the response rate to the family survey.

¹⁶ Out of all students in the private sector, 8.69% have no reported test scores, versus 4.82% of students in the public sector. While this difference cannot be explained by a reluctance on the part of private schools to make their students take the test (as it is obligatory for all students), it could be due to less consistent reporting of these scores to the Ministry of Education (Moulin 2022a).

¹⁷ This system is virtually non-existent in private schools. In my sample, which is restricted to non-switchers, 0.57% of students in private schools were enrolled in a non-academic track in the sixth grade, compared to 2.9% of students from public schools.

Table 1. Summary statistics.

	Boys			Girls		
	Treatment group	Control group	p-value	Treatment group	Control group	p-value
<i>Sociodemographic characteristics</i>						
Mother's occupational status						
Farmer	3.64%	1.46%	0.000	4.16%	1.41%	0.000
Self-employed	4.50%	3.07%	0.019	7.57%	2.97%	0.000
Managerial, professional and higher-level intellectual occupations	11.11%	6.86%	0.000	8.89%	6.48%	0.005
Intermediate occupations	24.04%	17.42%	0.000	22.33%	16.88%	0.000
Clerical and sales workers	39.46%	43.14%	0.029	40.96%	43.63%	0.111
Manual workers	9.58%	12.85%	0.003	8.04%	13.76%	0.000
Retired, economically inactive or NR	7.66%	15.20%	0.000	8.04%	14.89%	0.000
Father's occupational status						
Farmer	5.56%	2.66%	0.000	6.91%	2.44%	0.000
Self-employed	16.00%	8.44%	0.000	14.19%	9.01%	0.000
Managerial, professional and higher-level intellectual occupations	26.15%	15.80%	0.000	23.27%	14.32%	0.000
Intermediate occupations	18.49%	18.68%	0.883	16.18%	17.72%	0.227
Clerical and sales workers	9.20%	11.04%	0.079	10.69%	11.47%	0.466
Manual workers	22.41%	38.61%	0.000	25.35%	39.91%	0.000
Retired, economically inactive or NR	2.20%	4.76%	0.000	3.41%	5.12%	0.017
Mother's educational attainment						
Tertiary education	27.68%	17.34%	0.000	25.73%	16.54%	0.000
General upper secondary education	9.77%	7.95%	0.052	9.37%	7.15%	0.013
Technical or vocational upper secondary education	7.76%	6.08%	0.043	7.10%	6.21%	0.283
Vocational education	24.62%	23.66%	0.508	27.06%	24.31%	0.059
No qualification or NR	30.17%	44.97%	0.000	30.75%	45.79%	0.000
Father's educational attainment						
Tertiary education	26.44%	15.98%	0.000	22.80%	15.07%	0.000
General upper secondary education	5.46%	4.18%	0.066	4.82%	4.15%	0.320
Technical or vocational upper secondary education	7.28%	6.45%	0.327	6.05%	5.97%	0.915
Vocational education	31.13%	28.54%	0.092	32.83%	29.34%	0.024
No qualification or NR	29.69%	44.85%	0.000	33.49%	45.47%	0.000
Family type						
Two parents	88.22%	83.71%	0.000	85.81%	82.03%	0.003
Single mother	7.95%	11.45%	0.001	10.50%	12.60%	0.058
Single father	1.44%	1.34%	0.808	1.14%	0.96%	0.593
Blended family	2.11%	2.76%	0.231	2.08%	3.57%	0.014
Other situations (including NR)	0.29%	0.74%	0.101	0.47%	0.84%	0.211
Parents' countries of birth						
France	89.75%	77.41%	0.000	90.35%	78.47%	0.000
France/Maghreb	2.01%	3.83%	0.004	1.80%	3.44%	0.005
France/sub-Saharan Africa	0.29%	0.64%	0.170	0.66%	0.71%	0.856
France/other countries	2.97%	4.04%	0.101	2.55%	4.28%	0.009
Maghreb	0.96%	6.86%	0.000	1.14%	6.35%	0.000
Sub-Saharan Africa	0.38%	1.01%	0.051	0.38%	1.18%	0.019
Asia and Oceania	0.29%	1.32%	0.004	0.57%	0.84%	0.355
Turkey	0.19%	0.86%	0.023	0.09%	0.68%	0.023
European countries	1.72%	1.69%	0.940	1.42%	2.03%	0.189
Other countries or NR	1.44%	2.33%	0.071	1.04%	2.03%	0.030
Sibship size >3	45.59%	47.14%	0.361	43.42%	48.96%	0.001
<i>School characteristics</i>						
Size of the urban area						
Rural	5.75%	8.61%	0.002	6.24%	8.97%	0.004
< 5,000 inh	6.80%	9.39%	0.008	6.91%	10.12%	0.001
[5,000; 10,000] inh	10.06%	8.67%	0.151	10.12%	8.95%	0.229
[10,000; 20,000] inh	7.95%	6.84%	0.202	8.89%	7.21%	0.057
[20,000; 50,000] inh	10.54%	9.16%	0.163	10.03%	9.78%	0.805
[50,000; 100,000] inh	8.91%	7.13%	0.046	8.51%	7.15%	0.122
[100,000; 200,000] inh	7.66%	7.17%	0.578	7.85%	7.81%	0.962
[200,000; 2,000,000] inh	24.90%	19.87%	0.000	24.98%	19.28%	0.000
Paris agglomeration	11.59%	14.87%	0.006	13.43%	14.59%	0.330
Change of school locality	5.84%	8.28%	0.008	3.03%	6.14%	0.000
<i>Student's educational characteristics</i>						
Enrolled in a private primary school	67.53%	6.01%	0.000	68.12%	5.65%	0.000
Grade retention in primary school	10.82%	13.16%	0.039	7.66%	11.36%	0.000
6th-grade test score in mathematics (out of 100)	70.88	67.44	0.000	70.45	66.35	0.000
6th-grade test score in French (out of 100)	70.22	66.84	0.000	74.98	72.20	0.000
Number of observations	1044	5144		1057	5327	

The values reported are the sample means (except for test scores); p-values are for t-tests of equal means.

educational characteristics. The father of 26.15% (23.27%) of boys (girls) in the treatment group was in a managerial, professional, or higher-level intellectual occupation, versus 15.80% (14.32%) in the control group. Conversely, the father of 25.35% (22.41%) of girls (boys) in the treatment group was a manual worker, versus 39.91% (38.61%) in the control group. In terms of educational attainment, the mother of 27.68% (25.73%) of boys (girls) in the treatment group held a higher education qualification, versus 17.34% (16.54%) of boys in the control group. Whether for boys or girls, more public school than private school students had a single mother. The proportion of public school students whose parents were born abroad was also higher. Group differences in the size of the urban area reflect differences in the locations of private and public schools, which influence the likelihood of a student being enrolled in a private school (Moulin 2022a). The largest difference between students enrolled in private and public lower secondary schools was the type of elementary school they attended: 67.53% (68.12%) of the boys (girls) who were enrolled in a private lower secondary school had previously been enrolled in a private (or private and public) primary school, versus 6.01% (5.65%) of the boys (girls) enrolled in a public lower secondary school. Previous experience of grade repetition in elementary school was also more common among students in public schools. Finally, private school students had higher levels of knowledge in mathematics and French at the end of primary education, as measured by sixth-grade test scores.

Table 2 presents summary statistics for my dependent variables for boys and girls enrolled in private and public schools. The summary statistics show significant differences for dependent variables. For both boys and girls, the educational outcomes of students enrolled in a private lower secondary school differed from those of students enrolled in a public school. In comparison to public school students, private school students were more likely to obtain the *baccalauréat* and a higher education qualification. Finally, ten years after the *baccalauréat*, former private school students' wages were higher than those of former public school students. However, a simple comparison of outcomes between students enrolled in private schools and students in public schools does not measure the impact of private schooling. To identify a potential causal effect, I use propensity score matching, as detailed in the next Section.

IV. Empirical approach

Estimation strategy

The goal of my analysis is to estimate the causal impact of private lower secondary school attendance on a range of medium-term outcomes, both educational and non-educational, using propensity score matching.¹⁸ I rely on the potential outcomes framework originally proposed by Roy (1951) and Rubin (1974). The parameter of interest is the average treatment effect on the treated (ATT), defined as follows:

$$\tau_{ATT} = E[Y(1)|D = 1] - E[Y(0)|D = 1]$$

where $Y(1)$ and $Y(0)$ are potential outcomes with and without treatment D (equal to 1 for students who attended private lower secondary school and 0

Table 2. Summary statistics for dependent variables.

	Boys		Girls	
	Treatment group	Control group	Treatment group	Control group
<i>Educational outcomes</i>				
Obtaining the <i>baccalauréat</i>	31.23%	26.56%	48.16%	39.01%
Obtaining a higher education qualification	62.16%	53.71%	72.55%	62.78%
<i>Labour market outcomes</i>				
Monthly wage (in euros)	1897	1840	1638	1542

The values reported are the sample means (except for monthly wages).

¹⁸The implementation of the propensity score matching procedure follows Caliendo and Künn (2011), Caliendo, Künn, and Weißenberger (2016), Caliendo and Tübbicke (2020).

for students who attended public lower secondary school). $E[Y(0)|D = 1]$, the hypothetical outcome without treatment for students who received the treatment, is unobservable. This fundamental evaluation problem implies that this term should be derived from data on non-participants. Simple comparison of means between public and private school students will bias estimates. This is due to non-random assignment to the treatment and control groups, resulting from differences in observable characteristics X and unobserved characteristics U .

Propensity score matching methods aim to correct for this selection bias by balancing observable characteristics across the treatment and control groups (Rosenbaum and Rubin 1983). To give consistent estimates of the ATT, the conditional independence assumption (CIA) must hold. This implies that conditional on observed characteristics, the unobserved potential outcome is independent of treatment assignment:

$$Y(0) \perp\!\!\!\perp D | P(X)$$

where $P(X) = \Pr(D = 1|X)$ is the propensity score. This implies that the specification of the propensity score must include all characteristics that simultaneously influence the probability of treatment assignment and the outcome of interest. Therefore, ideally I will include in my framework all covariates related to private school attendance as well as the chosen educational outcomes, and labour market outcomes. To do this, I take advantage of an extensive database including sociodemographic characteristics, school characteristics, and students' educational characteristics. Moreover, the variables included in the specification must not themselves be influenced by the treatment. I also assume the overlap assumption: $0 < \Pr(D = 1|X) < 1$, X which implies that all students have a positive probability of being treated and of not being treated.

Assuming that these assumptions hold, I can estimate the ATT as the simple mean difference in outcome between treated and non-treated students on the sample appropriately re-weighted by the students' propensity score distribution (Caliendo and Kopeinig 2008):

$$\tau_{ATT} = E[Y(1)|P(X), D = 1] - E[Y(0)|P(X), D = 0]$$

If there is some unobserved covariate that has an effect on both private school enrolment and the outcomes, the CIA will fail. In this case, a hidden bias might arise, and matching estimators are no longer robust (Caliendo and Kopeinig 2008). Then private school attendance can be specified as:

$$\Pr(D = 1|X, U) = F(\beta X + \gamma U)$$

where $F(\cdot)$ is the logistic distribution. The study is free of hidden bias if $\gamma = 0$, whereas if $\gamma \neq 0$, two students with the same observed characteristics would differ in their odds of attending a private school. No hidden bias from unobserved confounders is a key assumption (Zhang 2017). In section 5.2 I use a sensitivity analysis to assess the influence of potential unobserved heterogeneity on my results.

Taking into account the possible endogeneity of school choice

The unconfoundedness assumption implies that the treatment (enrolment in a private school) is random conditional on observed characteristics. To ensure that this condition is met, a model that explains how students are assigned to private schools is needed. In France, to be enrolled in a private school, it is first necessary that the student's parents wish to enrol them, and second that the student be accepted into the private school. Let us analyse these two conditions in turn.

Firstly, the choice to send a child to private school in France depends on several determinants which have been highlighted by previous research. Notably, the probability of enrolling in a private school depends strongly on social background (Héran 1996) – which I take into account in the modelling by integrating socio-occupational category and parent's educational attainment –; having been enrolled in a private elementary school (Bertola 2017); initial academic level (Moulin 2022a, 2022b); and the private school's location in relation to the parents' home (Fougère et al. 2017) – which I include in the modelling via the size of the

urban unit, indirectly taking into account geographical disparities in the distribution of private schools.^{19,20,21}

Second, as mentioned in [section 3.1](#), admission to private school is subject to the selection of students on the basis of their initial academic level and their motivation to enrol in a private school. For the first, I include the student's grade academic level at the end of primary education as measured by sixth grade test scores in the modelling of the probability of attending a private school.²² I also include a variable indicating whether the student repeated a grade in elementary school, which also provides information about the students' academic level. Private schools also select students on their motivation to enter, which is assessed through a cover letter and/or an interview. While it is not possible to measure the motivation of students or their parents directly, work in the sociology of education highlights that it tends to be privileged families that possess the social dispositions (ways of being, thinking, communicating and acting) the closest to those of schools (Blanchard and Cayouette-Remblière 2016). Thus, these families are better able to draw on such resources to identify the expectations of private schools and maximize their children's chances of being admitted. Here I take the family's social origin, as measured by the parents' socio-occupational category and educational attainment, as a proxy for the student's motivation and her parents from the school's perspective in the selection process. In this motivational assessment strategy, the head of the school may also take into account the fact of the student having been previously enrolled in a private school in primary education, so this variable is also considered in the analysis.

Accounting for the determinants of student enrolment in private schools in the modelling means that, conditional on the propensity score, each student has the same probability of assignment to treatment, as in a randomized experiment. In this way, I control for the possible endogeneity of school choice.

Implementation of the matching procedure

Estimation of the propensity score

To ensure that the CIA holds, the modelling specification must also include variables that affect the outcomes being measured. I include variables related to the student's sociodemographic (parents' occupational status, parents' educational attainment, family type, parents' countries of birth and number of siblings) and academic characteristics (enrolment in a private primary school, grade retention in primary school, and sixth grade test scores in mathematics and French). This choice is guided by previous research: one on short-term school outcomes in the French context (Moulin 2022a), and one on educational attainment and earnings Sass et al. (2016), using students' demographics and educational characteristics in a matching approach.

The logit estimates of the propensity scores are presented in [Table 3](#). Several variables strongly impact the probability of being enrolled in a private school. In particular, boys whose fathers' occupational status was below the level of Managerial, professional and higher-level intellectual occupations were less likely to be enrolled in a private school (in contrast with students with a self-employed parent). Similarly, students one or both of whose parents were born in the Maghreb were less likely to be enrolled in a private school. Students enrolled in a school in a small urban area were less likely to be enrolled in a private school. This is partly related to the uneven supply of private schools across the country (see Moulin 2022a). The variable most strongly associated with the probability of being enrolled in a private lower secondary school is, unsurprisingly, having been enrolled in a private primary school.

The distribution of estimated propensity scores for the treated and control groups can be found in [Figure 1](#). A common support condition is imposed in order to compare students with identical propensity scores. Private school students are matched with public school students using a range of matching

¹⁹Tavan (2004) shows that private school student recruitment differs depending on the scale of the private sector in the local school market.

²⁰Vandenbergh and Robin (2004) adopt a similar strategy, using the location of the school in an urban area with more than 100,000 inhabitants as an instrumental variable, to the extent that more private school places are available in cities that exceed this threshold.

²¹In addition, according to the work of Bertola (2017) there does not seem to be *a priori* a link between higher socioeconomic status and the location of private schools. The distribution of private schools is higher in areas with a Catholic tradition (Bertola 2017; Tavan 2004).

²²These are standardized tests that all students take at the beginning of the sixth-grade school year and these assessments are among the few comparative measures of school attainment not influenced by the social and educational environment (Cayouette-Remblière and Moulin 2019).

Table 3. Logit estimation of the propensity scores.

	Boys	Girls
<i>Sociodemographic characteristics</i>		
Mother's occupational status		
Farmer	0.390	0.794*
Self-employed	-0.208	1.013***
Managerial, professional and higher-level intellectual occupations (ref)	-	-
Intermediate occupations	-0.002	0.236
Clerical and sales workers	-0.225	0.042
Manual workers	-0.286	-0.112
Retired, economically inactive or NR	-0.398	-0.017
Father's occupational status		
Farmer	0.200	0.549
Self-employed	0.393**	0.203
Managerial, professional and higher-level intellectual occupations (ref)	-	-
Intermediate occupations	-0.361**	-0.309*
Clerical and sales workers	-0.374*	-0.217
Manual workers	-0.423**	-0.192
Retired, economically inactive or NR	-0.708**	-0.156
Mother's educational attainment		
Tertiary education (ref)	-	-
General upper secondary education	-0.034	-0.070
Technical or vocational upper secondary education	0.119	0.009
Vocational education	-0.043	0.105
No qualification or NR	-0.103	-0.100
Father's educational attainment		
Tertiary education (ref)	-	-
General upper secondary education	0.023	0.290
Technical or vocational upper secondary education	-0.011	-0.168
Vocational education	0.245	0.074
No qualification or NR	0.051	0.086
Family type		
Two parents (ref)	-	-
Single mother	-0.036	-0.158
Single father	0.708*	0.573
Blended family	-0.519	-0.097
Other situations (including NR)	-0.312	-0.304
Parents' countries of birth		
France (ref)	-	-
France/Maghreb	-0.600***	-0.547*
France/sub-Saharan Africa	-1.060	-0.074
France/other countries	-0.625**	-0.527*
Maghreb	-1.079***	-1.177***
Sub-Saharan Africa	0.046	-0.536
Asia and Oceania	-1.184*	-0.261
Turkey	-0.403	-0.852
European countries	0.282	-0.420
Other countries or NR	-0.069	-0.735
Sibship size >3	0.028	-0.183*
<i>School characteristics</i>		
Size of the urban area		
Rural	-0.836***	-0.970***
< 5, 000 inh	-0.445**	-0.915***
[5, 000; 10, 000] inh	-0.172	-0.369*
[10, 000; 20, 000] inh	-0.083	-0.108
[20, 000; 50, 000] inh	0.064	-0.083
[50, 000; 100, 000] inh	0.414**	0.287
[100, 000; 200, 000] inh	0.199	-0.067
[200, 000; 2, 000, 000] inh (ref)	-	-
Paris agglomeration	0.395**	0.211
Change of school locality	-0.645***	-1.346***
<i>Student's educational characteristics</i>		
Enrolled in a private primary school	3.487***	3.642***
Grade retention in primary school	0.018	-0.253
6th-grade test score in mathematics	0.003	0.013***
6th-grade test score in French	-0.001	-0.013***
Constant	-0.985	-0.659
Observations	6188	6384

Reported coefficients are derived from a logit model. Standard errors are in parentheses.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

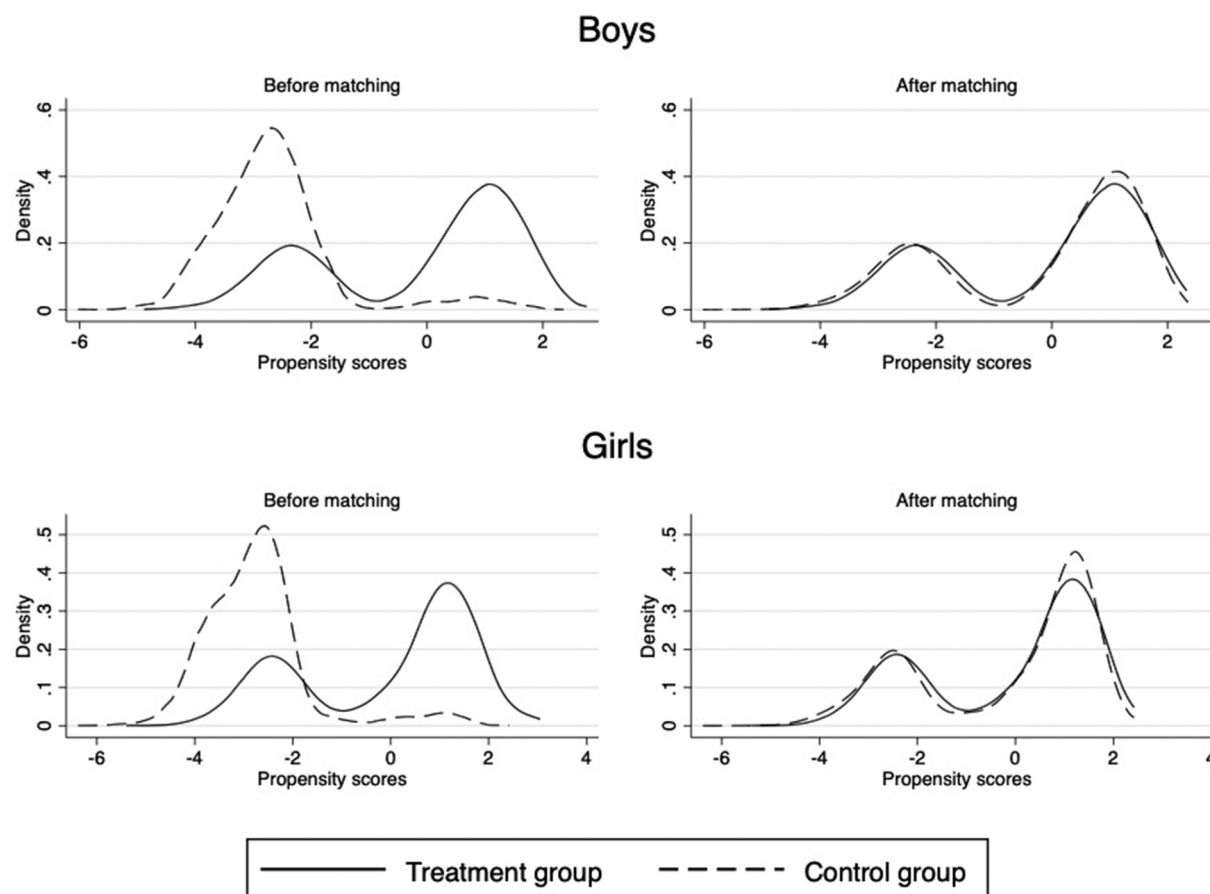


Figure 1. Propensity score distribution. *Note:* This figure displays the propensity score distributions for the treatment and control group before (left side) and after (right side) the matching procedure.

estimators (nearest neighbour without replacement, kernels with different bandwidths, radius and inverse probability weighting).

To account for attrition, I use specific weights for each set of outcomes in my estimates. For each outcome, I follow the method described in DuGoff, Schuler, and Stuart (2014) and then I include corresponding weights as predictors in the propensity score before multiplying the propensity score weights by the survey sampling weights.

Matching quality

Table 4 presents several commonly used indicators of matching quality. These balance indicators before and after matching are reproduced for the different weights (one for each survey) that I use in my estimation strategy. After matching there are almost no differences in means at the 1% significance level. In terms of mean absolute standardized bias, all

of my specifications are below the threshold of 3–5% suggested by Caliendo and Kopeinig (2008), confirming the high quality of the matching procedure. In addition, pseudo- R^2 for propensity score estimates are very low after matching, and the p -value of joint significance tests is equal to 1 after matching for all specifications. The matching procedures thus eliminate the difference in observable characteristics between students enrolled in private and public schools.

V. Estimation results

Main results

Table 5 presents the estimated average treatment effects of attending a private lower secondary school on medium-term educational and labour market outcomes.

Table 4. Matching quality indicators.

	Family survey				EVA survey			
	Boys		Girls		Boys		Girls	
	Before matching	After matching	Before matching	After matching	Before matching	After matching	Before matching	After matching
Number of variables with significant differences in means								
At the 1% level	21	0	22	0	14	0	15	0
At the 5% level	27	0	29	1	19	0	24	0
At the 10% level	33	4	32	1	22	0	25	1
Mean absolute standardized bias in %	14.11	2.79	14.44	2.18	12.22	3.02	12.73	2.89
Median absolute standardized bias in %	7.50	2.32	9.15	2.00	7.35	2.66	8.67	2.50
Pseudo-R ² for propensity score estimation	0.3707	0.0081	0.3889	0.0076	0.3797	0.0119	0.4057	0.0140
P-value of joint significance test	0.0000	0.9984	0.0000	0.9994	0.0000	0.9998	0.0000	0.9903
Total number of variables	47	47	47	47	46	46	45	45
Treated students off support		3		21		9		20

I use Epanechnikov kernel matching with a bandwidth of 0.06 to calculate matching quality indicators.

* $p \leq 0.1$; ** $p \leq 0.05$; *** $p \leq 0.01$.

Table 5. Matching estimation results.

	Obtaining the <i>baccalauréat</i>	Obtaining a higher education qualification	Monthly wage (log)
<i>Boys</i>			
Nearest neighbour (1:1) with replacement	0.048*** (0.001)	0.051*** (0.003)	0.050*** (0.003)
Nearest neighbour (1:5) with replacement	0.024*** (0.001)	0.079*** (0.003)	0.034*** (0.003)
Kernel 1 (bw = 0.02)	0.017*** (0.001)	0.082*** (0.003)	0.041*** (0.003)
Kernel 2 (bw = 0.06)	0.026*** (0.001)	0.086*** (0.003)	0.047*** (0.003)
Kernel 3 (bw = 0.2)	0.045*** (0.001)	0.096*** (0.003)	0.062*** (0.003)
Radius (caliper = 0.1)	0.040*** (0.001)	0.090*** (0.003)	0.059*** (0.003)
Inverse probability weighting	0.017*** (0.001)	0.056*** (0.004)	0.050*** (0.003)
Standard errors			
Number of observations	6188	3139	2469
<i>Girls</i>			
Nearest neighbour (1:1) with replacement	0.061*** (0.001)	0.060*** (0.003)	0.021*** (0.003)
Nearest neighbour (1:5) with replacement	0.061*** (0.001)	0.068*** (0.003)	0.037*** (0.003)
Kernel 1 (bw = 0.02)	0.068*** (0.001)	0.071*** (0.003)	0.044*** (0.003)
Kernel 2 (bw = 0.06)	0.068*** (0.001)	0.063*** (0.003)	0.050*** (0.003)
Kernel 3 (bw = 0.2)	0.077*** (0.001)	0.079*** (0.003)	0.061*** (0.003)
Radius (caliper = 0.1)	0.073*** (0.001)	0.073*** (0.003)	0.059*** (0.003)
Inverse probability weighting	0.059*** (0.001)	0.016*** (0.004)	0.041*** (0.005)
Standard errors			
Number of observations	6384	3662	2846

The following variables are included in the logit estimation when estimating the propensity score: mother's occupational status, father's occupational status, mother's educational attainment, father's educational attainment, family type, parents' countries of birth, size of the urban area, enrolment in a private primary school, grade retention in primary school, sixth-grade test score in mathematics, sixth-grade test score in French. The logit estimates of the propensity scores are presented in Table 3.

Standard errors are in parentheses.

* $p \leq 0.1$; ** $p \leq 0.05$; *** $p \leq 0.01$.

In terms of educational outcomes, my results show that enrolling in a private school has a positive impact on the likelihood of obtaining the *baccalauréat*. This effect appears to be greater for girls as it is between 0.059 and 0.077 SD whereas it is between 0.017 and 0.048 SD for boys. In both cases, the seven estimators give positive and significant results. In comparison, these effects are much smaller than those that can be observed for short-term outcomes. Indeed, in assessing the effect of private schooling on educational outcomes at the end of secondary school, Moulin (2022a) shows that boys' scores in private school were between 0.193 and 0.222 SD higher on standardized tests in ninth grade than those of public school students. This effect is in the range of 0.138–0.198 for girls enrolled in a private lower secondary school.

Moreover, private school enrolment had a positive effect on the likelihood of obtaining

a higher education qualification both for boys and girls (between 0.051 and 0.096 SD for boys and 0.016 and 0.079 for girls).

In terms of labour market outcomes, my estimates highlight a positive effect on wages. Thus, on average, a boy who was enrolled in a private lower secondary school will earn between 3.4% and 6.2% more than a student who was enrolled in a public school. This effect is similar for girls who were enrolled in a private lower secondary school, ranging from 2.1% to 6.1%. While the raw monthly wage differences are larger for girls than for boys (see Table 2), after matching the differences are quite similar.

Robustness analysis

The robustness of these findings on the impact of private schooling on educational and labour market outcomes must be assessed. I check the

robustness of my main results in two ways. First, using different estimation methods, I assess the sensitivity of the estimates. Second, I check the robustness of the estimated treatment effects with respect to deviation from the identifying assumption. If an unobserved variable affects the probability of belonging to the treatment group then the CIA does not hold and there is a hidden bias. Although it is not possible to directly test the validity of the CIA, I use sensitivity analysis. The analysis simulates a potential confounder to assess the robustness of my estimates with respect to deviation from the CIA.

The first robustness check is presented above in Table 5: it consists of using different estimators to estimate the effects of private schooling in order to ensure that the results do not depend on the choice of estimator. My results are totally stable across these different specifications, implying that they are not sensitive to the choice of algorithm.

My second robustness check is based on a simulation approach (Nannicini 2007; Ichino, Mealli, and Nannicini 2008). This method assesses the influence of potential unobserved heterogeneity. The sensitivity analysis is based on the central assumption that the CIA holds,

i.e. that observed characteristics X and unobserved components U that affect the outcomes and the probability of assignment to the treatment group are fully integrated into the matching procedure. The principle of this analysis is to simulate the impact of an unobserved variable U on the outcomes of interest under the assumption that its distribution is similar to that of an observable characteristic X . The method enables a precise evaluation of the influence of the observed variable on outcomes and treatment selection, and links it directly to the potential unobserved leverage effect (Caliendo and Künn 2011). Table 6 presents the results of this simulation-based sensitivity analysis. The outcome effect is given as the average estimated odds ratio of the unobserved component U in the logit model $\Pr(Y = 1|D = 0, X, U)$. The selection effect is the average estimated odds ratio of U in the logit model $\Pr(D = 1|X, U)$.

For all the outcomes, the coefficients are of a similar intensity to the main results. This means that none of the simulated confounders would alter the estimated effects of private school attendance on educational and labour market outcomes. These results suggest that the estimated treatment effects

Table 6. Sensitivity to unobserved heterogeneity – simulation approach.

	Obtaining the baccalaureate			Obtaining a HE qualification			Monthly wage		
	Out. effect	Sel. effect	ATT	Out. effect	Sel. effect	ATT	Out. effect	Sel. effect	ATT
<i>Boys</i>									
No confounder			0.026			0.086			0.047
Mother's socio-professional category – Clerical and sales workers	0.690	0.897	0.037	0.869	0.823	0.080	0.684	0.831	0.050
Father's socio-professional category – Manual workers	0.359	0.484	0.021	0.332	0.542	0.066	0.939	0.484	0.048
Mother's educational attainment – Tertiary education	4.917	1.678	0.027	3.909	1.542	0.075	1.211	1.764	0.050
Father's educational attainment – Tertiary education	4.421	1.718	0.028	4.229	1.682	0.073	1.240	1.824	0.050
Family types – Two parents	1.974	1.405	0.035	1.877	1.427	0.079	0.754	1.566	0.052
Parents' countries of birth – France	1.363	2.634	0.031	1.404	2.568	0.074	0.483	2.512	0.054
Size of the urban area – Paris agglomeration	1.024	0.736	0.038	1.244	0.757	0.081	2.026	0.784	0.052
Enrolled in a private primary school	0.896	34.092	0.051	0.971	39.869	0.098	0.877	37.004	0.058
Grade retention in primary school	0.119	0.895	0.036	0.137	1.024	0.080	1.183	0.867	0.051
<i>Girls</i>									
No confounder			0.068			0.063			0.050
Mother's socio-professional category – Clerical and sales workers	0.750	0.920	0.070	0.849	0.883	0.081	0.902	0.884	0.057
Father's socio-professional category – Manual workers	0.404	0.533	0.055	0.432	0.568	0.070	0.876	0.513	0.053
Mother's educational attainment – Tertiary education	4.047	1.643	0.062	4.325	1.459	0.075	1.370	1.771	0.054
Father's educational attainment – Tertiary education	4.176	1.571	0.063	5.010	1.443	0.075	1.668	1.714	0.055
Family types – Two parents	1.825	1.312	0.068	1.585	1.138	0.081	0.658	1.278	0.057
Parents' countries of birth – France	1.182	2.628	0.066	1.220	2.683	0.078	0.573	2.724	0.060
Size of the urban area – Paris agglomeration	0.994	0.892	0.071	1.480	0.905	0.082	2.135	0.952	0.057
Enrolled in a private primary school	0.883	38.110	0.092	1.106	45.949	0.063	1.044	42.552	0.049
Grade retention in primary school	0.135	0.706	0.066	0.132	0.833	0.078	1.225	0.683	0.057

Average treatment effects on the treated (ATT) are calculated using kernel propensity-score matching estimates. The estimation is based on 200 replications. Source: 2007 panel of secondary school students in France, MEN-DEPP.

seem not to be sensitive to unobserved heterogeneity, which confirms the robustness of my baseline estimates.

VI. Discussion and conclusion

In this paper, I have estimated the effect of private lower secondary schooling on a range of outcomes during adulthood using propensity score matching. Using data from a rich system of surveys, the present study goes beyond the question of the effects of private school attendance on academic achievement, analysing effects on a range of educational and labour market outcomes until seventeen years after students enter the sixth grade. It is the first study to assess the effect of private schooling on medium-term outcomes in France. In terms of the effects of private schooling on educational outcomes in the years after lower secondary schooling, I have shown that private school attendance increases the likelihood of obtaining the *baccalauréat* and a higher education qualification for both boys and girls. In terms of labour market outcomes, I find a positive and significant effect of private schooling on boys and girls' earnings at age 27. The quality of the matching procedure is very good, and a set of robustness checks confirms the robustness of these results.

Exploring the factors that may explain the results of my study is all the more important because in France, the institutional settings of private and public schools are very similar (Valdenaire 2011). Private schools operating under government contract are strictly regulated by the state, and receive public funds. Various elements of private and public schools are similar: the required teaching qualifications are the same, teachers' salaries are paid by the state, and the curriculum is identical. The use of propensity score matching and the robustness of the results allow me to exclude the endogeneity of parents' choice of private school as an explanatory factor. The question of what does explain the results remains an open one. In the remainder of this section, I discuss possible explanations for this phenomenon.

First, the positive effects observed in this article may originate from the way in which the private schools operate. The first explanation concerns a feature distinctive to private schools: the

discretionary teacher recruitment by the head of the school. This way of recruiting teachers may allow for greater team cohesion around the school's educational project, which may in turn have an impact on the quality of education provided to students. Indeed, the school's educational project defines the conditions under which the national educational programs are implemented in the school. This project must specify the school's academic and extracurricular activities and must indicate the resources employed to ensure the success of all students and involve parents in this objective. We can legitimately assume, as Moulin (2022a) does, that when teachers are freely recruited by the head of the school, they gather more around the school's educational project than when they are assigned administratively, like teachers in public schools. It is this greater involvement of teachers in the school's educational project that can explain which may in turn have an impact on the quality of education provided to students. Another explanation is the possible difference in interaction between students and teachers in private schools. In public schools, teachers are assigned to a school, whereas in private schools teachers choose their school (they choose to apply and are then recruited or not by the head of the school). As a result, private school teachers have more choice about the professional environment in which they teach and may therefore be more able to invest in their work, which may potentially result in different interactions between students and teachers. These interactions can then potentially explain the positive effects observed in terms of medium term educational outcomes. Finally, a further explanation lies in the possible difference in the forms of management between the heads of schools in public schools and those in private schools. Indeed, management by the head of the school can have positive effects on student achievement. For example, Tavares (2015) shows that the implementation of specific management practices (such as performance monitoring, target setting and incentive systems) in Brazil has improved student achievement. In the same vein, using a school-level randomized field experiment in Texas, Fryer Roland (2017) shows that offering management training to principals can significantly increase student achievement. It cannot be excluded that forms of

management are more efficient in private schools, resulting for example from the Catholic tradition of these schools in France, which would have a positive impact on student outcomes.

Second, the positive effects observed on boys and girls in this paper can be traced back to the peers of private schools students. This may lie firstly in the social composition of schools and classes. Although the presented estimates of the effect of private schools are based on controlled social characteristics, the overall composition of the schools and classes is not included in our estimates due to lack of available data. However, this social composition²³ is more favourable for private schools (see Table 1), that may affect students' academic success by spurring mutual emulation and raising the overall academic level, and then explain the medium term effects of private schooling. Moreover, it is also possible that peer effects in a broader sense [including peers' backgrounds, current behaviour, or outcomes that affect an outcome as in the definition of Sacerdote 2011) are stronger in private schools in comparison to public schools. The fact that students are automatically assigned to the local public school, whereas in private schools parents choose the school in which to enrol their children, may potentially lead to differences in the peer effects of these two types of school and then positively affecting the academic performance of students in these private schools.

The possible explanations I have raised in the discussion, which cannot be verified directly due to lack of specific data, are, however, consistent with the literature on the medium-term effects of charter schools in the United States. This literature explains that medium-term gains from charter school attendance can be explained by the skills that these schools provide to students (Dobbie and Fryer 2015; Sass et al. 2016). Thus, private schools are likely to provide students skills that lead to short-term gains (Moulin 2022a) which are then sustained over time.²⁴

My results call for a better understanding of the mechanisms that explain the greater efficiency of private schools in France. Indeed, once these mechanisms are isolated, it will be possible to replicate what works in private schools and apply it to

public schools, which would benefit to students in these schools. Fryer (2014) shows that injecting best practices from charter schools into public schools in Texas increases student maths achievement. It is therefore possible to imagine doing the same in France.

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ORCID

Léonard Moulin  <http://orcid.org/0000-0001-6144-9678>

Data availability statement

Data are available to researchers via Quetelet Progedo Diffusion.

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²³The social composition of private schools is a factor in explaining the better results of students in these schools (see Dronkers and Rober 2003, 2008).

²⁴With regard specifically to the labour market, beyond the better skills that private schooling confers on students, which are valued through higher wages, it cannot be excluded that private school students, who are aware of their better educational performance, have higher reservation wages than public school students. This hypothesis could be pursued in further research.

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